

ATs for Sight

05332/05632/17332/17732 Fall 2026

Introduction to Accessibility and Assistive Technology

<https://cmu-intro-to-accessibility.github.io>

Andrew Begel

Early Course Feedback (15 min)

- Please help me improve this class by taking this anonymous survey.



<https://bit.ly/ecf-05332>

Administrivia

- P2 was due at 11am today.
- P3 is due Friday, October 9, 2026, 11:59pm.
- HW4 due Wednesday 11am.
- Homework pace will slow down as the project ramps up.
- Accessibility Lunch today:
 - Akshay Nayak: **Improving Data Chart Usability for Low-Vision Users**

HW1 Review: Accessible PDFs

- Tables
 - Need to mark headers with <TH> (only the top row, if it's really a header)
 - Need to mark rows with <TR>
 - Need to mark table cells with <TD> (unless it's a header, then <TH>)
- Lists
 - Need to mark each list as <L>
 - Need to mark each list item as
- Sometimes the PDF *has* accessibility tags but thinks it does not.
 - Rerun the tagger in Adobe Acrobat.

HW1 Review: Accessible Slides

1. High contrast should include **all** screen elements.
2. Decorative images need not have alt-text.
3. Emojis are a clever way to indicate secondary notation because they are their own alt-text.
4. Progressive display should include animating bullets when there's a whole bunch on the screen.
5. You can add text overlays to graphics with embedded text or put the words in the alt-text.
6. Watch out for colorblindness effects.
 - Not just red-green, but blue-yellow and black-and-white vision.
7. Unless you are the original author, do not materially change the slide content or order.

HW1: Accessible Presentations

- I wanted you to give the talk, not explain how you made it accessible.
 - No points deducted: Will fix this in HW1 for next year.
- Remember who your majority audience is.
 - When pointing out slide elements for blind and visually impaired viewers, don't hurt the talk for sighted viewers.
 - Don't make the talk longer. No one gets an extended talk slot for accessibility.
- When explaining a chart, say the takeaway message first, then point out only the details in the chart necessary to understand it.

HW1: Accessible Presentations

- Pick a single location for an embedded talking head.
 - Moving the location every slide can be very distracting for people with cognitive differences.
- When a picture is an adjunct to a story, focus on telling the story.
 - Definitely describe the picture, but make the description fit *into* your story. It shouldn't come afterwards like a footnote.

Project 3: Project Specifications

Brainstorming Learning Goals

- Customers ask for things but don't always realize what they need.
- Your goal is to help them envision what you could do to meet those unwritten needs.
- P2 was one-sided, on purpose.
- This let you do due diligence on the organization, its mission, its audience, and the idea.
- This gives you the ability to appear knowledgeable when you meet your partner organization in P3.

P2 Investigation

1. Who is my partner organization?
2. From what I can tell already,
 - What are the project requirements?
 - Who is the audience?
 - What exactly do they want me to create?
 - How do I break this project down into smaller parts?
3. What else do I need to find out from my partner?
4. You created three possible ideas for the project, inspired by the partner organization's expressed idea yet bounded by your team's skillset and creativity.

Negotiation Learning Goals

- Work with your partner organization to come up with a project plan that you both can agree on.
- This is a negotiation, not a hostage situation.
 - They proposed one idea.
 - You will counter with three possible visions of how to accomplish the idea.
 - Gauge their reactions. Were you headed in the right direction?
 - What pieces of your three ideas did they like?
 - Combine the best parts into a better solution!

Project Requirements

- Once you've nailed down the project idea with your partner organization, make it concrete enough to work on.
 - **Stakeholders** – Who are your users?
 - **Value Proposition** – What problem are you solving?
 - **Technology** – What tech do you need to accomplish it?
 - **Deliverables** – What does your partner org want at the end?
 - **Project Stages** - You've got 3 2-week milestones in which to do this project. How will you divide up the work across the semester?

User Stories

“As a [specific user], I want to [action], so that [benefit]”

- Plus acceptance criteria:
 - How do we know when we're done?

Example User Story

- Google Chrome Web Browser
- As a [CMU student], I want to [quickly find my course home page] so I can [remind myself when the project is due].
- Acceptance criteria:
 - A CMU student can get to the course web page and find their assignment due date in less than 10 seconds.

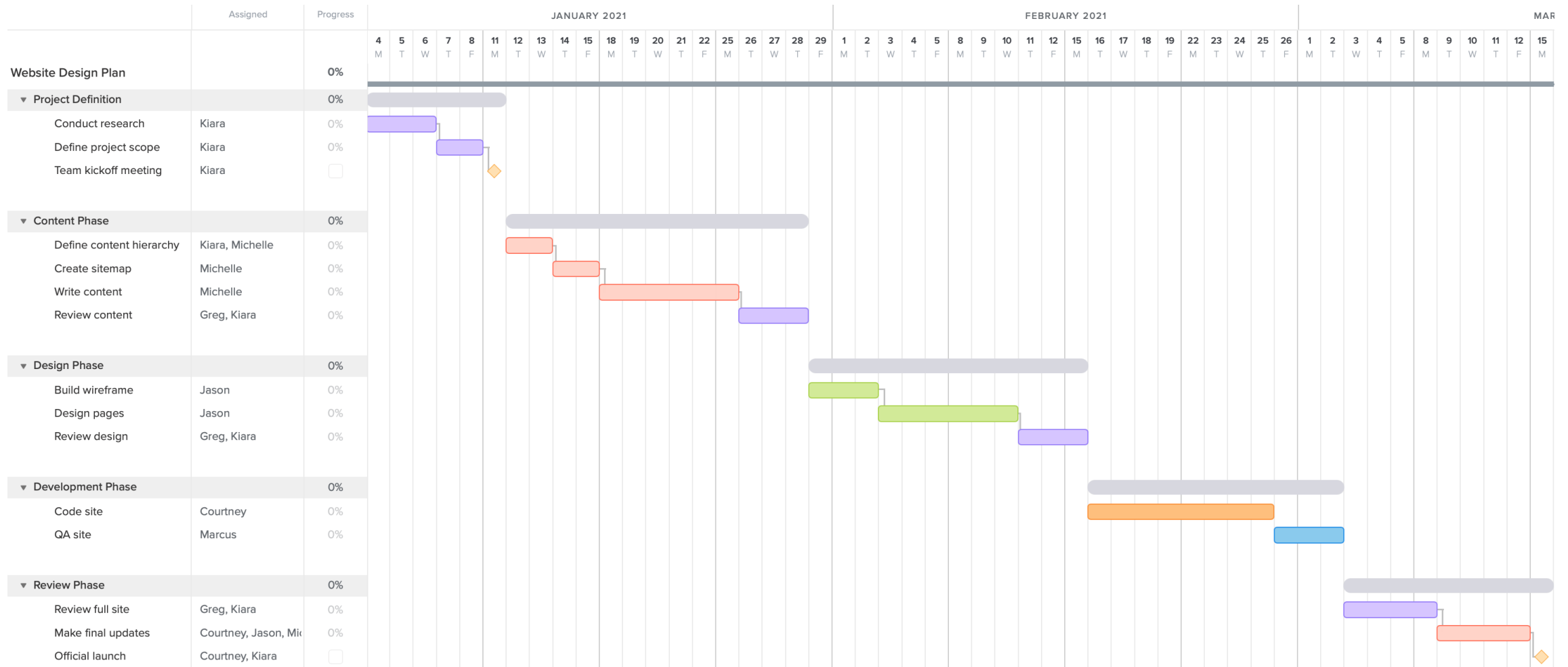
User Stories

- Create 8 or 10 user stories.
 - 8 user stories for a 4-person team.
 - 10 user stories for a 5-person team.
- Put your user stories in priority order.
 - Impact – How essential is this to your stakeholders?
 - Effort – How much time/effort is required to do it?
- Allocate each user story to Milestone 1, 2, or 3.
- Assign each user story primary and secondary owners.

Human Subjects Review

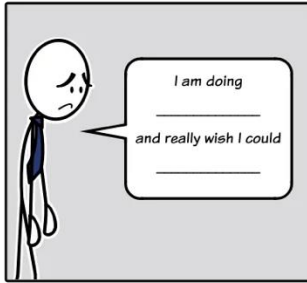
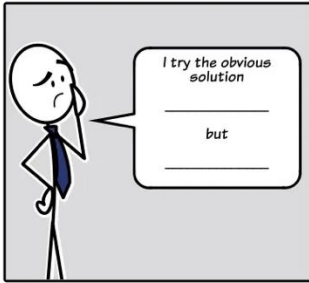
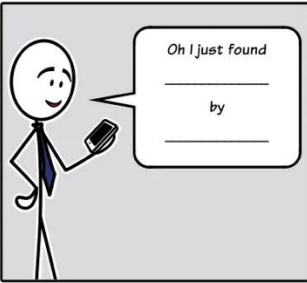
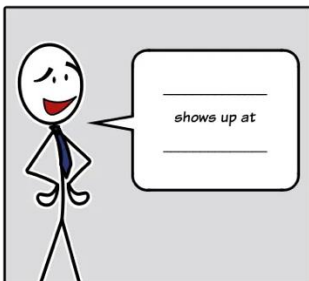
- You will test your project with real users.
 - Other members of the class
 - People with the disability you are targeting.
- All user studies in USA are governed by a set of laws to protect the people in the study from the researchers.
- CMU's IRB requires all researchers to undergo review.
 - Fill out a study protocol to explain what you will do to whom.
 - Create an informed consent form for your users.

Gantt Chart



Storyboards

- Create UI mockups, storyboards, wireframes, etc.
- This will help your partner organization understand what you're going to do for them.

Problem Experienced	Solution Search	Product Discovery
		
Clearly show an example scenario where the customer is experiencing a pain point where they would want your product.	For most problems there is an existing solution, explain why it doesn't quite work in your scenario.	Show a plausible way your user could be introduced to your product/offering. Google? Word of Mouth?
Product Experienced	Problem Alleviated	Beneficial Outcome
		
Show a magical solution that largely ignores the bigger question of how the technology works behind the scene.	After the magic button is pressed, what happens from a customer perspective?	Why is the customer happy? What benefit to them was experienced?

Create your own at Storyboard That

Assistive Technologies for Sight

Glasses

According to glasses.com, the first pair of glasses as we know them for correcting vision were invented sometime around the 13th century in Italy.



Text to Speech

Uses Synthesized voices to generate audio or speech output of text on the screen

First speech synthesizer was Bell Telephone Laboratory's Voder. Invented by Homer Dudley in 1937.

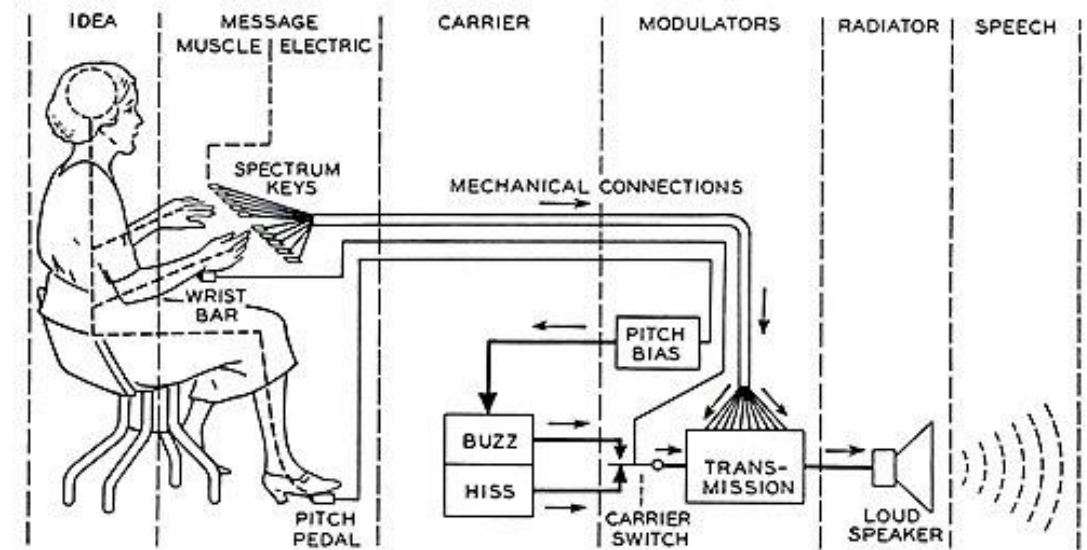


Fig. 8—Schematic circuit of the voder.

Screen Readers

Reads the words on the screen

Keystroke to move to next area

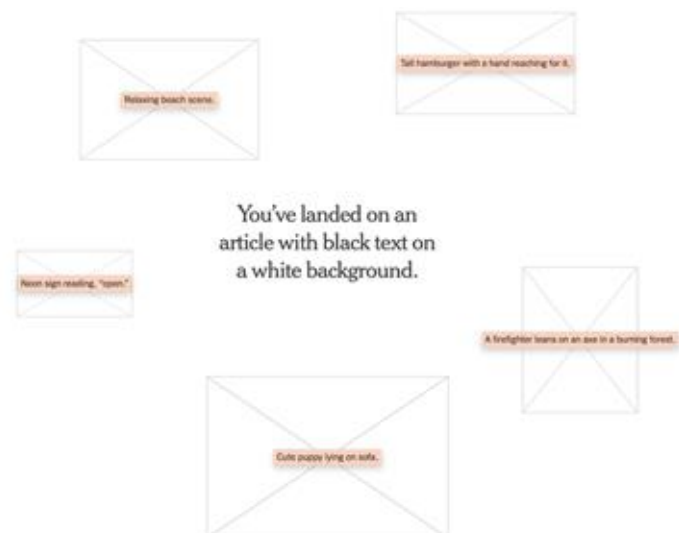
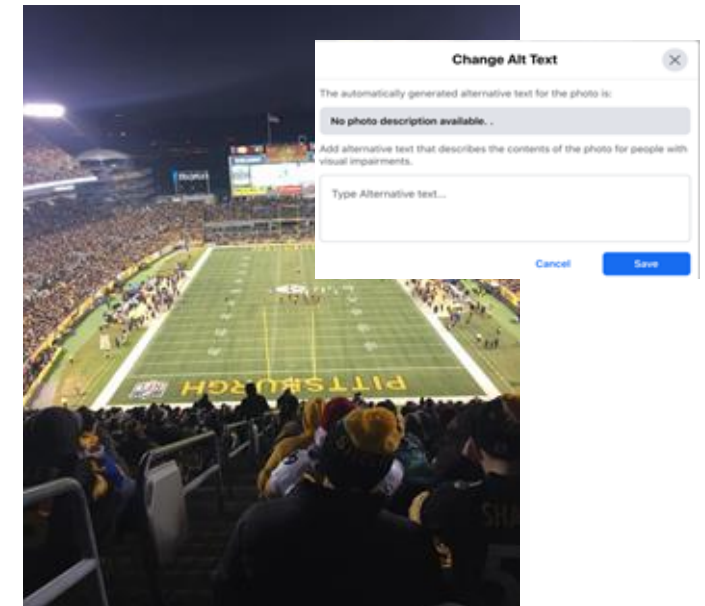
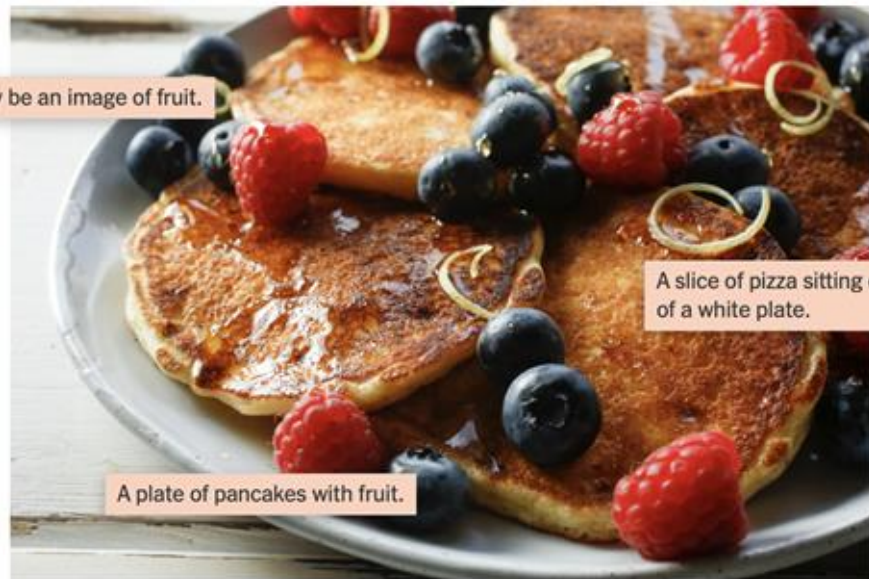
Blind people can operate them amazingly quickly

Examples of Screen Readers:

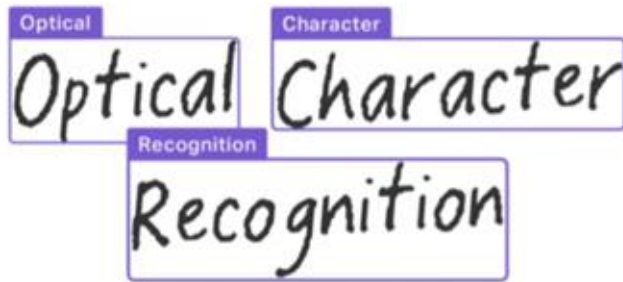
JAWS, NVDA, VoiceOver, TalkBack

Alternative Text

Alternative text provides a way to add text descriptions to image elements



Optical Character Recognition (OCR)



Extracts information from handwritten or printed text.

OCR software can automatically extract data from printed documents, saving money and time for processing them.

Screen Magnification

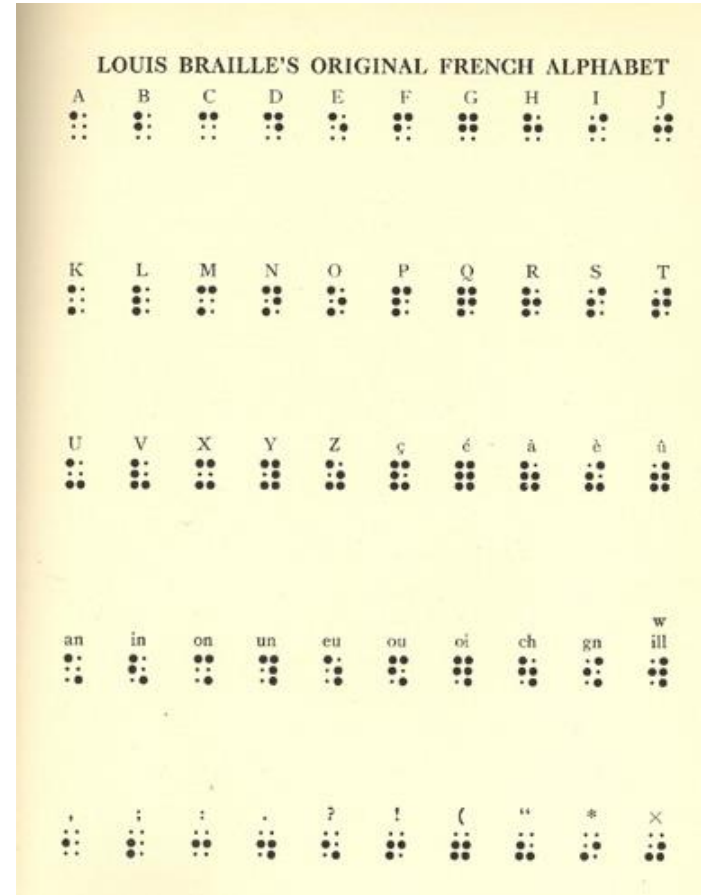
Screen magnification apps are used by individuals with low vision. These apps take a computer or mobile device's visual output and enlarge it to allow for easier and more detailed viewing.

Braille

Braille Code invented by Louis Braille in the 1800s. French Alphabet.

Uniform Braille code not adopted in English-Speaking countries until 1932

Embossed dots. 6 or 8 dot cells



Refreshable Braille Displays

Braille displays provide access to information on a computer screen by electronically raising and lowering different combinations of pins in braille cells

Many have about 40 cells.



Braille Note Takers

More Similar to a Personal Digital Assistant

Read/write files, emails, music, basic web browsing



Humanware BrailleNote Touch

White Cane

People have been using sticks to aid walking for thousands of years.

White cane invented in the early 1900s.

Very lightweight, multiple tips.

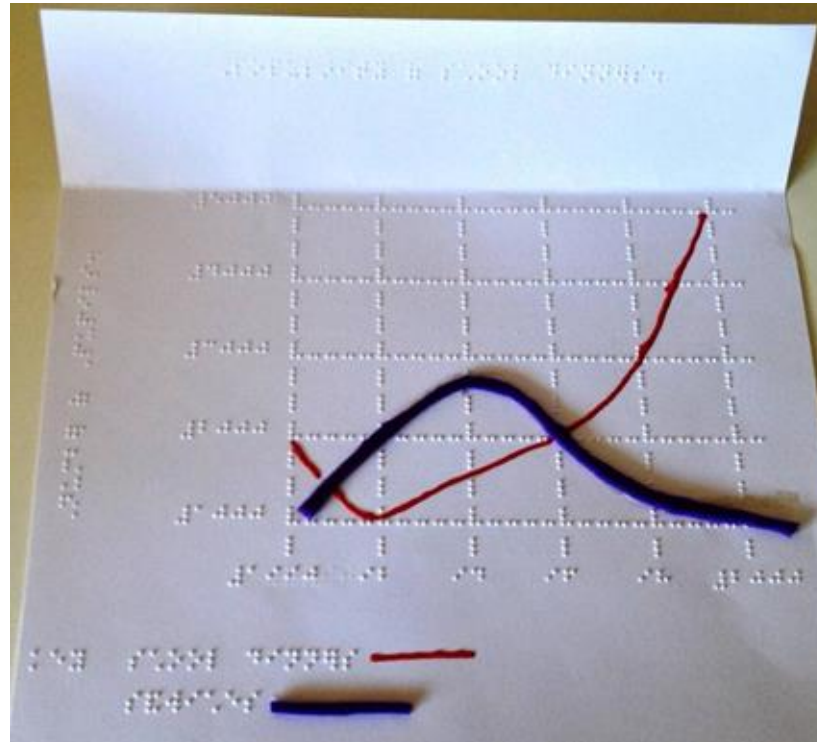


Guide Dogs / Seeing Eye Dogs

- 150,000 years ago, we domesticated dogs
- First training of dogs to assist blind people late 1700s
- 1920-1930s we see guide dog schools in the UK



Tactile Signs and Graphics



GPS Navigation Tools

1980s military applications

~1996 when GPS becomes
available for civilians

Connecting to Mapping Software
and Applications



VizWiz

VizWiz

Introduction VizWiz Crowd Agents Scribe

VizWiz



Bigham et al. Nearly Real-Time Answers to Visual Questions. UIST 2010.

University of Rochester Human-Computer Interaction

Jeffrey P. Bigham



CaneTroller

Enabling People with Visual Impairments to Navigate Virtual Reality with a Haptic and Auditory Cane Simulation

Yuhang Zhao^{1,2}, Cynthia L. Bennett^{1,3}, Hrvoje Benko¹, Edward Cutrell¹, Christian Holz¹,
Meredith Ringel Morris¹, Mike Sinclair³

¹Microsoft Research, Redmond [benko, cutrell, cholz, merrie, sinclair]@microsoft.com
²Information Science, Cornell Tech, Cornell University yz769@cornell.edu
³Computer Science & Engineering, University of Washington benne3@uw.edu



Figure 1. (A) A blind user wearing the gear for our VR evaluation, including a VR headset and CaneTroller, our haptic VR controller. (B) The mechanical elements of CaneTroller. (C) Overlays of the virtual scene atop the real scene show how the virtual cane extends past the tip of the CaneTroller device and can interact with the virtual trash bin. (D) The use of CaneTroller to navigate a virtual street crossing: the inset shows the physical environment, while the rendered image shows the corresponding virtual scene. Note that users did not have any visual feedback when using our VR system. The renderings are shown here for clarity.

ABSTRACT

Traditional virtual reality (VR) mainly focuses on visual feedback, which is not accessible for people with visual impairments. We created CaneTroller, a haptic cane controller that simulates white cane interactions, enabling people with visual impairments to navigate a virtual environment by transferring their cane skills into the virtual world. CaneTroller provides three types of feedback: (1) physical resistance generated by a wearable programmable brake mechanism that physically impedes the controller when the virtual cane comes in contact with a virtual object; (2) vibrotactile feedback that simulates the vibrations when a cane hits an object or touches and drags across various surfaces; and (3) spatial 3D auditory feedback simulating the sound of real-world cane interactions. We designed indoor and outdoor VR scenes to evaluate the effectiveness of our controller. Our study showed that CaneTroller was a promising tool that enabled visually impaired participants to navigate different virtual spaces. We discuss potential applications supported by CaneTroller ranging from entertainment to mobility training.

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Author Keywords

Virtual reality; white cane; blindness; visual impairments; haptic feedback; auditory feedback; mobility.

ACM Classification Keywords

H.5.1. Information interfaces and presentation: Multimedia Information Systems. K.4.2. Computers and Society: Assistive technologies for persons with disabilities.

INTRODUCTION

Virtual reality (VR) technologies are maturing quickly and have been widely applied to different fields by researchers and designers, such as entertainment [6,8], education [32,54], and social activities [82]. Researchers have recognized VR's potential in accessibility and developed various VR systems for training and rehabilitation for people with different disabilities [25], such as dyslexia [17,21], stroke [14,33], and ADHD [1,34].

While presenting great potential in a variety of use cases, current VR solutions rely mainly on realistic visual feedback to provide an immersive experience to sighted people, for whom the visual sense is dominant [38]. Most VR applications are not accessible for people with visual impairments (VIPs), preventing them from benefiting from this important class of emerging technologies [28].

Some prior research has explored VR experiences for VIPs. However, they either generated pure auditory virtual reality that has limited capability in describing the shape or specific layout of a virtual space [28,36], or provided

